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The Editors

*npj Materials Degradation*

Nature Portfolio

Dear Editors,

We are pleased to submit our manuscript, “**Sparse data limit what a corrosion model can predict,**” for consideration as an Article in the Collection *Computational Modeling of Corrosion*.

The Collection asks for work that moves corrosion research from phenomenological description toward mechanism-based prediction. This manuscript addresses a precondition for that move which the field has largely left untested: whether the data a mechanistic model is calibrated against actually determine its parameters. Mechanistic corrosion models are routinely fitted with four to six parameters to a handful of measurements and then extrapolated across service lifetimes. We show, for a representative case, that several parameters are not determined, and that the consequences are large enough to change a lifetime assessment. The case is the duplex oxide grown on Nb-stabilized AISI 347 (X6CrNiNb18-10) in simulated boiling-water-reactor hydrothermal water (240 °C, 7 MPa, 0.4 ppm dissolved O<sub>2</sub>), modeled with a reduced point defect model (PDM) and identified by differentiable, physics-informed inversion. Five aspects speak directly to the Collection’s aims.

- **A transferable computational framework.** Profile likelihood, a 1000-member bootstrap, a prior-leverage test and a residual-leverage decomposition are built into the inversion rather than bolted on afterward, made affordable by exact parameter gradients through a vectorized solver. They show that of the five parameters of the accepted model, three exposure times leave one undetermined and pin a second only to within an order of magnitude, and that 95% of the fit statistic is carried by three chromium points, 80% by a single measurement. These tools are mature in systems biology and battery modeling but, to our knowledge, have not been brought to the PDM. The diagnosis transfers to any sparsely sampled degradation model, which is most of them.
- **Corrosion prognosis and service-life assessment, stated at the resolution the data support.** The identified kinetics fit the calibration data as well as an empirical power law but extrapolate differently: at ten years the power law lies a factor of 3.5 to 6.3 above the mechanistic barrier layer and outside its uncertainty band under every error treatment we apply, including one that widens that band from 447–704 nm to roughly 340–1040 nm. We are deliberate that the *separation* is the result and the point estimate it brackets (535 nm) is not. An analysis whose thesis is that sparse data under-determine parameters should not then quote a lifetime number as though it were determined. A designed exposure of roughly 3000 hours would resolve the separation. That is a concrete, costed experiment for service-life assessment, and the reason a re-analysis of data already in the literature is worth doing.
- **A failure mode in physics-informed corrosion modeling, diagnosed and remedied.**

Physics-informed learning is being adopted quickly in this field. We document a failure specific to sparse experimental data, in which the data loss is satisfied while the governing equation is not, so a trajectory that looks like an excellent fit reports parameters that are wrong. We remedy it with a forward-solve self-consistency check that we propose for reporting alongside any physics-informed inverse result. Because the reduced kinetics have a closed form, the framework can be graded against an exact reference rather than trusted, which is what makes the diagnosis possible.

- **Mechanistic insight at the electrochemical interface, including a negative result.** Extended spatially with no additional free parameters, the identified kinetics predict the measured composition profile, and a structural argument shows why composition data cannot constrain the defect diffusivity however many are taken. Solving Poisson’s equation self-consistently with the defect transport, instead of imposing the field, shows that the PDM’s constant-field closure is not supported at the defect concentrations the model itself implies: the implied Debye length is about 0.06 nm, several times smaller than an interatomic spacing. This leaves the fitted kinetics untouched, since the identification is of  $b_3$  and not of the field, but it bounds how derived field strengths may be read.
- **Passivation modeling for an alloy and environment that had none.** Prior characterization of AISI 347 oxide in BWR water stopped at empirical per-element power-law fits with no shared physical parameter set. We connect the Cr-rich barrier and Fe-rich outer layers through a single kinetic description built on a constant-volume constraint, and it is that coupling, not fit quality, that drives the divergent extrapolations. We are equally explicit about the limits: the model structure is not itself Nb-specific, and the neural inversion is validated against an exact classical reference rather than required by the problem.

The manuscript is original, has not been published elsewhere, and is not under consideration by any other journal. The author declares no competing interests. The work involves no human or animal subjects. All code, the digitized data, and a committed artifact behind every number in the manuscript are available in a public repository, together with a generated map from each figure and table to the command that produces it, so that any reported value can be re-derived without re-running the full pipeline.

Thank you for considering our submission.

Sincerely,

Conrard Giresse Tetsassi Feugmo